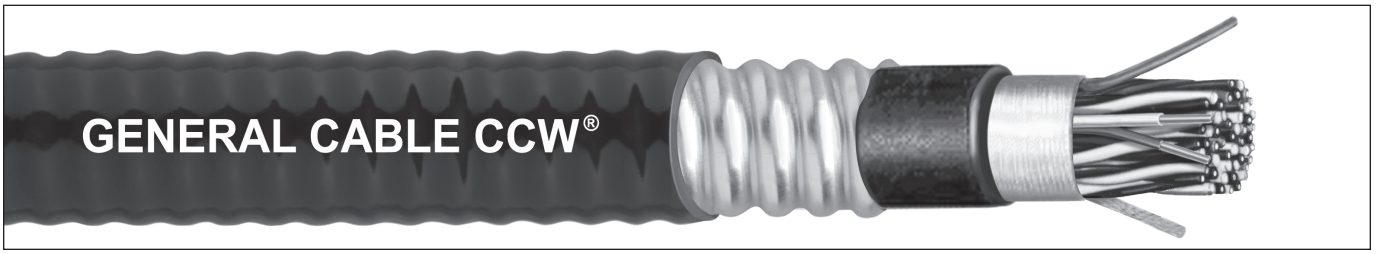
CCW® Armored Instrumentation, Pairs/Triads, Overall Shield

UL Type ITC-HL/PLTC, PVC, 300 V, 105°C, Sunlight-Resistant, Direct Burial
UL Marine Shipboard Cable, ABS CWCMC



Product Construction:

Conductor:
- Bare annealed copper per ASTM B3
- Class B stranding per ASTM B8

Insulation:
- Flame-retardant Polyvinyl Chloride (PVC), rated 105°C per UL Standards 13 and 2250
- Color-coded per ICEA Method 1: pairs – black and white; triads – black, white and red. Each conductor in each pair or triad is printed alphanumerically for easy identification

Cable Assembly:
- Individual pairs or triads and communication wire are cabled together with a left-hand lay
- Communication wire: 22 AWG solid bare copper, flame-retardant Polyvinyl Chloride (PVC), rated 105°C, orange
- Communication wire is not included on single pair or single triad cables

Overall Shield:
- Flexfoil® aluminum/polyester tape shield providing 100% coverage
- Stranded tinned copper drain wire, same size as insulated conductors

Inner Jacket:
- Flame-retardant Polyvinyl Chloride (PVC) per UL Standards 13 and 2250, black
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C
- Nylon rip cord to facilitate jacket removal

CCW Armor:
- Impervious, continuously welded and corrugated aluminum alloy sheath per UL 1569
- CCW armor conductivity meets the grounding requirements of NEC Article 250

Overall Jacket:
- Flame-retardant, moisture- and sunlight-resistant Polyvinyl Chloride (PVC) per UL Standards 13 and 2250, black
- Low temperature performance meets ASTM D746 brittleness temperature at or below -40°C

Applications:

- CCW armored Instrumentation cables with an overall shield provide superior protection and reliability against physical damage for use in instrumentation and process control applications requiring ITC-HL or PLTC wiring methods where shielding against external EMI is required
- For use as Power Limited Tray Cable on circuits rated 150 V or less and 5 amps or less in Class 2 or Class 3 circuits in accordance with NEC Article 725
- For use as Instrumentation Tray Cable on circuits rated 150 V or less and 5 amps or less in accordance with NEC Article 727
- Recognized for use in Class I, II, and III, Divisions 1 and 2; or Class I, Zones 1 and 2 hazardous locations per NEC Articles 501, 502, 503, and 505
- Installed indoors or outdoors, in wet or dry locations, in a raceway, as aerial cable on a messenger, in cable trays, or for direct burial
- Recognized for use on fixed or floating offshore petroleum facilities as recommended by the American Petroleum Institute

Features:

- CCW armor provides superior mechanical protection and an impervious barrier to moisture, gas and liquids
- CCW armor provides EMI shielding performance
- Meets cold impact at -40°

Specifications:

Design Adherence:
- UL 13 Power-Limited Circuit Cables
- UL 2250 Instrumentation Tray Cable
- UL 1569 Metal Clad Cables
- UL 1309/CSA C22.2 No. 245 Marine Shipboard Cable

Flame Tests:
- ICEA T-29-520 (210,000 BTU/hr)
- IEEE 383 (70,000 BTU/hr)
- CSA FT4
- IEEE 1202 (70,000 BTU/hr)
- UL 1581 (70,000 BTU/hr)
- IEC 60332-3 Cat. A

Compliances:
- UL Type PLTC, SUN RES, DIR BUR, -40°C, UL File # E36118
- UL Type ITC-HL, UL File # E177408
- UL Listed Marine Shipboard, UL File # E85994
- American Bureau of Shipping (ABS) Listed for CWCMC

General Cable
Phone: 888-593-3355
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CCW® Armored Instrumentation, Pairs/Triads, Overall Shield

UL Type ITC-HL/PLTC, PVC, 300 V, 105°C, Sunlight-Resistant, Direct Burial
UL Marine Shipboard Cable, ABS CWCMC

PART NUMBER	COND. SIZE (AWG)	NO. OF PAIRS	INSULATION THICKNESS		COMMUNICATION WIRE			INNER JACKET THICKNESS		NOMINAL CORE O.D.		NOMINAL ARMOR O.D.		JACKET THICKNESS		NOMINAL OVERALL O.D.		CROSS-SECTIONAL AREA[1]	APPROXIMATE NET WEIGHT	
					SIZE	INS. THICKNESS														
			mils	mm	AWG	mils	mm	mils	mm	INCHES	mm	INCHES	mm	mils	mm	INCHES	mm	(SQ. IN.)	LBS/1000 FT	kg/1000 m
20 AWG 7W (0.52 mm²) OVERALL SHIELDED PAIRS																				
9225.20021221	20	2	15	0.38	22	12	0.30	40	1.02	0.35	8.9	0.58	14.7	50	1.27	0.69	17.5	0.38	174	260
9225.20041221	20	4	15	0.38	22	12	0.30	50	1.27	0.41	10.4	0.62	15.7	50	1.27	0.73	18.5	0.42	198	294
9225.20061221	20	6	15	0.38	22	12	0.30	50	1.27	0.45	11.4	0.67	17.0	50	1.27	0.78	19.8	0.48	250	372
9225.20081221	20	8	15	0.38	22	12	0.30	50	1.27	0.49	12.4	0.71	18.0	50	1.27	0.82	20.8	0.54	283	420
9225.20101221	20	10	15	0.38	22	12	0.30	50	1.27	0.53	13.5	0.75	19.1	50	1.27	0.86	21.8	0.59	330	492
9225.20121221	20	12	15	0.38	22	12	0.30	50	1.27	0.56	14.2	0.80	20.3	50	1.27	0.91	23.1	0.66	353	525
9225.20161221	20	16	15	0.38	22	12	0.30	60	1.52	0.66	16.8	0.89	22.6	50	1.27	1.00	25.4	0.80	432	644
9225.20201221	20	20	15	0.38	22	12	0.30	60	1.52	0.73	18.5	0.97	24.6	50	1.27	1.08	27.4	0.93	500	744
9225.20241221	20	24	15	0.38	22	12	0.30	60	1.52	0.79	20.1	1.06	26.9	50	1.27	1.17	29.7	1.09	556	827
9225.20361221	20	36	15	0.38	22	12	0.30	70	1.78	0.94	23.9	1.24	31.5	50	1.27	1.35	34.3	1.45	711	1,058
9225.20501221	20	50	15	0.38	22	12	0.30	70	1.78	1.06	26.9	1.34	34.0	50	1.27	1.45	36.8	1.67	949	1,412
18 AWG 7W (0.82 mm²) OVERALL SHIELDED PAIRS																				
9225.18021221	18	2	15	0.38	22	12	0.30	40	1.02	0.37	9.4	0.58	14.7	50	1.27	0.69	17.5	0.38	206	307
9225.18041221	18	4	15	0.38	22	12	0.30	50	1.27	0.44	11.2	0.67	17.0	50	1.27	0.78	19.8	0.48	249	370
9225.18061221	18	6	15	0.38	22	12	0.30	50	1.27	0.50	12.7	0.71	18.0	50	1.27	0.82	20.8	0.54	294	437
9225.18081221	18	8	15	0.38	22	12	0.30	50	1.27	0.55	14.0	0.80	20.3	50	1.27	0.91	23.1	0.66	337	501
9225.18101221	18	10	15	0.38	22	12	0.30	60	1.52	0.65	16.5	0.89	22.6	50	1.27	1.00	25.4	0.80	417	621
9225.18121221	18	12	15	0.38	22	12	0.30	60	1.52	0.68	17.3	0.93	23.6	50	1.27	1.04	26.4	0.86	449	668
9225.18161221	18	16	15	0.38	22	12	0.30	60	1.52	0.77	19.6	1.02	25.9	50	1.27	1.13	28.7	1.02	536	797
9225.18201221	18	20	15	0.38	22	12	0.30	60	1.52	0.83	21.1	1.06	26.9	50	1.27	1.17	29.7	1.09	623	927
9225.18241221	18	24	15	0.38	22	12	0.30	60	1.52	0.86	21.8	1.11	28.2	50	1.27	1.22	31.0	1.18	698	1,038
9225.18361221	18	36	15	0.38	22	12	0.30	70	1.78	0.99	25.1	1.29	32.8	50	1.27	1.40	35.6	1.56	965	1,436
9225.18501221	18	50	15	0.38	22	12	0.30	70	1.78	1.15	29.2	1.47	37.3	50	1.27	1.58	40.1	1.99	1,280	1,905
16 AWG 7W (1.31 mm²) OVERALL SHIELDED PAIRS																				
9225.16010001	16	1	15	0.38	–	–	–	35	0.89	0.26	6.6	0.43	10.9	50	1.27	0.54	13.7	0.23	135	201
9225.16021221	16	2	15	0.38	22	12	0.30	40	1.02	0.36	9.1	0.58	14.7	50	1.27	0.69	17.5	0.38	239	355
9225.16041221	16	4	15	0.38	22	12	0.30	50	1.27	0.48	12.2	0.71	18.0	50	1.27	0.82	20.8	0.54	296	440
9225.16061221	16	6	15	0.38	22	12	0.30	50	1.27	0.57	14.5	0.80	20.3	50	1.27	0.91	23.1	0.66	366	545
9225.16081221	16	8	15	0.38	22	12	0.30	60	1.52	0.63	16.0	0.84	21.3	50	1.27	0.95	24.1	0.72	424	631
9225.16101221	16	10	15	0.38	22	12	0.30	60	1.52	0.69	17.5	0.93	23.6	50	1.27	1.04	26.4	0.86	522	777
9225.16121221	16	12	15	0.38	22	12	0.30	60	1.52	0.74	18.8	0.97	24.6	50	1.27	1.08	27.4	0.93	570	849
9225.16161221	16	16	15	0.38	22	12	0.30	60	1.52	0.85	21.6	1.11	28.2	50	1.27	1.22	31.0	1.18	687	1,023
9225.16201221	16	20	15	0.38	22	12	0.30	70	1.78	0.96	24.4	1.24	31.5	50	1.27	1.35	34.3	1.45	849	1,263
9225.16241221	16	24	15	0.38	22	12	0.30	70	1.78	1.04	26.4	1.34	34.0	50	1.27	1.45	36.8	1.67	956	1,422
9225.16361221	16	36	15	0.38	22	12	0.30	70	1.78	1.20	30.5	1.51	38.4	60	1.52	1.65	41.9	2.17	1,283	1,910
9225.16501221	16	50	15	0.38	22	12	0.30	80	2.03	1.33	33.8	1.64	41.7	60	1.52	1.78	45.2	2.52	1,786	2,658

PART NUMBER	COND. SIZE (AWG)	NO. OF TRIADS	INSULATION THICKNESS		COMMUNICATION WIRE			INNER JACKET THICKNESS		NOMINAL CORE O.D.		NOMINAL ARMOR O.D.		JACKET THICKNESS		NOMINAL OVERALL O.D.		CROSS-SECTIONAL AREA[1]	APPROXIMATE NET WEIGHT	
					SIZE	INS. THICKNESS														
			mils	mm	AWG	mils	mm	mils	mm	INCHES	mm	INCHES	mm	mils	mm	INCHES	mm	(SQ. IN.)	LBS/1000 FT	kg/1000 m
20 AWG 7W (0.52 mm²) OVERALL SHIELDED TRIADS																				
9225.20041222	20	4	15	0.38	22	12	0.30	50	1.27	0.43	10.9	0.67	17.0	50	1.27	0.78	19.8	0.48	247	367
9225.20081222	20	8	15	0.38	22	12	0.30	50	1.27	0.52	13.2	0.75	19.1	50	1.27	0.86	21.8	0.59	336	499
9225.20121222	20	12	15	0.38	22	12	0.30	50	1.27	0.60	15.2	0.84	21.3	50	1.27	0.95	24.1	0.72	427	636
9225.20161222	20	16	15	0.38	22	12	0.30	60	1.52	0.70	17.8	0.93	23.6	50	1.27	1.04	26.4	0.86	528	786
9225.20241222	20	24	15	0.38	22	12	0.30	60	1.52	0.84	21.3	1.11	28.2	50	1.27	1.22	31.0	1.18	694	1,032
9225.20361222	20	36	15	0.38	22	12	0.30	70	1.78	1.00	25.4	1.29	32.8	50	1.27	1.40	35.6	1.56	955	1,421
18 AWG 7W (0.82 mm²) OVERALL SHIELDED TRIADS																				
9225.18041222	18	4	15	0.38	22	12	0.30	50	1.27	0.49	12.4	0.71	18.0	50	1.27	0.82	20.8	0.54	291	433
9225.18081222	18	8	15	0.38	22	12	0.30	50	1.27	0.59	15.0	0.80	20.3	50	1.27	0.91	23.1	0.66	409	609
9225.18121222	18	12	15	0.38	22	12	0.30	60	1.52	0.72	18.3	0.97	24.6	50	1.27	1.08	27.4	0.93	554	824
9225.18161222	18	16	15	0.38	22	12	0.30	60	1.52	0.82	20.8	1.06	26.9	50	1.27	1.17	29.7	1.09	671	999
9225.18241222	18	24	15	0.38	22	12	0.30	70	1.78	0.97	24.6	1.24	31.5	50	1.27	1.35	34.3	1.45	936	1,393
9225.18361222	18	36	15	0.38	22	12	0.30	70	1.78	1.06	26.9	1.34	34.0	50	1.27	1.45	36.8	1.67	1,256	1,869
16 AWG 7W (1.31 mm²) OVERALL SHIELDED TRIADS																				
9225.16010002	16	1	15	0.38	-	-	-	35	0.89	0.28	7.1	0.43	10.9	50	1.27	0.54	13.7	0.23	160	238
9225.16041222	16	4	15	0.38	22	12	0.30	50	1.27	0.51	13.0	0.75	19.1	50	1.27	0.86	21.8	0.58	353	525
9225.16081222	16	8	15	0.38	22	12	0.30	60	1.52	0.67	17.0	0.89	22.6	50	1.27	1.00	25.4	0.80	540	803
9225.16121222	16	12	15	0.38	22	12	0.30	60	1.52	0.79	20.1	1.06	26.9	50	1.27	1.17	29.7	1.09	727	1,082
9225.16161222	16	16	15	0.38	22	12	0.30	70	1.78	0.92	23.4	1.19	30.2	50	1.27	1.30	33.0	1.34	922	1,372
9225.16241222	16	24	15	0.38	22	12	0.30	70	1.78	1.11	28.2	1.42	36.1	50	1.27	1.53	38.9	1.86	1,254	1,866
9225.16361222	16	36	15	0.38	22	12	0.30	80	2.03	1.31	33.3	1.60	40.6	60	1.52	1.73	43.9	2.38	1,792	2,667

Dimensions and weights are nominal; subject to industry tolerances.
[1] Cross-sectional area for cable tray fill is in accordance with NEC® Section 392.22.

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